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1 Overview

baramundi Connect (abbr. bConnect) is a REST web server, that gives you the opportunity to request data objects (like endpoints, jobs, etc.) out of the baramundi Management Suite and to create new or modify data objects with the baramundi Management Suite (bMS). Only an application that is using a HTTP web client with a version 1.1 or above is needed for access.

The REST web server expects an authentication according to the *BasicAuthentication* method. The provided credentials as well as the data are transmitted securely using SSL/TLS encryption. Beside the authentication against the Active Directory (in domains) or local rights (in work groups) the credentials are also used for authorization in the baramundi Management Suite. The scope of operations supported by bConnect and the results are therefore limited to the configured rights of the specified user in the baramundi Management Suite.

JSON and XML are the two data formats supported by bConnect. They can be configured either over the URL or in the HTTP request header. If no format is specified, JSON is used as default.

2 Configuration

bConnect is listening on HTTPS standard port 443 for incoming requests. The port can be configured in the view Interfaces of module Configuration, if required. Therefore you need to switch to the edit mode by selecting the menu point Edit in the context menu or in the menu bar as shown in Figure 1. If a HTTPS port number different from the default is used, it has to be set explicitly in the request URL.

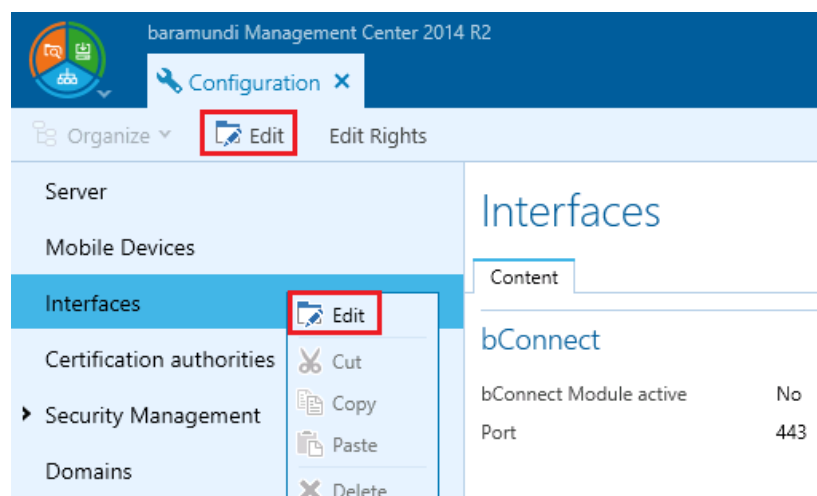


Figure 1 – Interface configuration

The same view provides the possibility to activate or deactivate the bConnect service. If Yes is configured for “bConnect Module active”, the service is listening to incoming requests. Otherwise it is deactivated. A restart of the baramundi Management Server (bServer) is necessary for changes to take effect.

As bConnect is using the HTTPS protocol by default for communication a valid server certificate needs to be bound to the configured port. By default the *baramundi Mobile Devices Module*, automatically creates a certificate for the baramundi Root Certification Authority (CA) and an appropriate server certificate – even no license for *baramundi Mobile Devices* is available. The server certificate is automatically bound to the standard HTTPS port 443. With default configuration bConnect and *baramundi Mobile Devices Module* share the same port. Since it is not possible to bind more than one certificate to a port, they also share the same server certificate. Hence no further settings are required when using default configuration.

Changing the communication port number for bConnect requires an appropriate binding of the certificate, which can be either done with Internet Information Server (IIS) or “netsh” on command line. The usage of an external server certificate, e.g. bought from public Certification Authority or self-created, requires a different port number and a new appropriate binding.

Some web clients or underlying libraries check the complete certificate chain by default. To avoid problems it is recommended to import the certificates of all CAs down to the Root CA to the Trusted Root Certification Authorities store of the machine, the web client is running on. The certificate of the baramundi Root CA used for *baramundi Mobile Devices* can be exported over the corresponding configuration site.

3 Calling Syntax

bConnect exposes over so called *Controller* several actions for requests. Each controller handles usually all requests concerning a specific object type. Therefore the bConnect API consist of several controllers where each of them provides specific actions according to the object type they represent.

The *Info* and *Version* controller provide general or version information of bConnect and can be requested with URLs of the following scheme:

```
https://<BMSServer>:<Port>/bConnect/<Controller>.
```

The other controller provide functionality to manage bMS objects (e.g. endpoint, job). The URL requests for this controller must include the bConnect version number:

```
https://<BMSServer>:<Port>/bConnect/v<bConnect Version>/<Controller>.
```

If the HTTPS standard port number 443 is configured for bConnect (default configuration), the port declaration is voluntary for the URL. Otherwise the port is mandatory.

bConnect supports only JSON and XML data format. If it is not possible to specify the data format in the HTTP-Header, e.g. because using a browser, it can be appended as an extension to the controller in the URL as follows:

```
https://<BMSServer>:<Port>/bConnect/<Controller>.<Format>
```

```
https://<BMSServer>/bConnect/v<bConnect Version>/<Controller>.<Format>
```

If XML format is used for POST, PUT and PATCH-Requests the elements must be sorted alphabetically and the following namespaces must be appended as attributes to the root element.

```
xmlns:i="http://www.w3.org/2001/XMLSchema-instance"
```

```
xmlns="http://schemas.datacontract.org/2004/07/Baramundi.Bms.Server.Connector.Rest.BConnect.DataModel"
```

Different controller actions can be requested by appropriate HTTP methods and additional parameter(s) in the URL. All parameters specified for a specific action are mandatory. A parameter is a key-value-pair connected over the "="-sign. They must be appended (without blanks) with a "?"-sign at the end of the base URL and in case of multiple they must be concatenated (without blanks) with a "&"-sign.

In case a specific resource is identifiable by its ID it can be addressed for all supported actions either by appending the ID as further segment in the URL or as parameter.

```
https://<BMSServer>:<Port>/bConnect/<Controller>.<Format>/<ID>?param1=1234
```

```
https://<BMSServer>:<Port>/bConnect/<Controller>.<Format>?id=<ID>&param2=4567
```

Further request examples (HTTP method "GET" only):

- <https://bmsserver.customer.net/bConnect/Info>
- <https://bmsserver.customer.net/bConnect/Version.xml>
- <https://bmsserver.customer.net/bConnect/v1.1/endpoints.json>
- <https://bmsserver.customer.net/bConnect/v1.1/jobinstances.xml/9d6a3f49-5c66-40c6-ae80-d6bd85fe3628> (equal to example below)
- <https://bmsserver.customer.net/bConnect/v1.1/jobinstances.xml?id=9d6a3f49-5c66-40c6-ae80-d6bd85fe3628> (equal to example above)
- <https://bmsserver.customer.net/bConnect/v1.1/jobinstances.json?endpointid=26f8284e-538d-422a-9b01-7597382bb71d&jobid=104196a1-1862-45fe-be3f-b312fcfc1ca9&startifexists=false>

4 Authentication & Headers

4.1 Authentication

To be able to use the bConnect module, it is mandatory to authenticate with each request your identity with the use of username and password. Both need to be specified in ISO-8859-1 encoding. Currently only authentication method *Basic Authentication* is supported. Therefore the credentials needs to be passed in the appropriate format in the designated HTTP request header or in the URL as mentioned above.

4.2 Headers

With each API call you can set specific request headers. With bMS 2016 two further headers are supported.

4.2.1 Request Header

Header	Description	Since bMS
Accept-Encoding	A standard HTTP request header, which contains a list of supported (compression) encodings by the web client. The encoding type of the response data is specified in the HTTP response header "Content-Encoding". Currently the (compression) encodings "gzip" and "deflate" are supported by bConnect.	2016
Client-Request-ID	A custom HTTP request header, which contains a unique ID (in general a GUID) generated by the requester. It will be returned unchanged in the header of the corresponding response. This header helps in certain scenarios to keep track on assignment of a request and corresponding response.	2016

4.2.2 Response Header

Header	Description	Since bMS
Content-Encoding	A standard HTTP response header, which contains the (compression) encoding type used on the response data. The encoding type is specified with the HTTP request header "Accept-Encoding". Currently the (compression) encodings "gzip" and "deflate" are supported by bConnect.	2016
Client-Request-ID	A custom HTTP response header, which contains a unique ID (in general a GUID) generated by the requester. It will be returned (unchanged) as received in the corresponding HTTP request header. This header helps in certain scenarios to keep track on assignment of a response and corresponding request.	2016

5 Controllers and actions

Each controller, like *Endpoints* or *Jobs*, is addressed with an explicit base URL. Using different HTTP methods or adding parameters to the URL requests different actions of the specified controller. Therefore different result is returned in the response. Controllers and parameters used in the URL are case insensitive.

The response data format is either configured over the .JSON or .XML extension in the URL (case insensitive) or over the *Accept*-field in the HTTP header. In latter case an extension must not be specified in the URL.

The response code of each request is a standard HTTP status code, e.g. one of the following:

- 200 → OK
- 201 → Created
- 204 → No Content
- 400 → Bad Request
- 401 → Unauthorized
- 403 → Forbidden
- 404 → Not Found
- 405 → Method Not Allowed
- 411 → Length Required
- 415 → Unsupported Media Type
- 500 → Internal Server Error
- 501 → Not Implemented
- 503 → Service Unavailable

baramundi software GmbH reserves the right to extend bConnect by new controllers, to extend existing controllers by new actions, to extend object types by new properties and to skip transmission of properties with value *null* without notification or change of the bConnect version number.

5.1 Controller Version

The *Version* controller provides actions concerning the *Version* object. *Version* is a bConnect internal object type which contains the current (newest) bConnect version and an array with all still supported versions.

To request the *Version* object the URL must not contain version information and the provided user does not need configured rights in the baramundi Management Suite (bMS).

URL: `https://<bmsserver>:<port>/bConnect/Version`

Method	URL parameters	Description	Since bMS
GET	None	Returns an object of type <i>Version</i> .	2014

[Properties of *Version*.](#)

Examples:

- GET-Request:
`https://bmsserver.customer.net/bConnect/version.json`

Result:

```
{
  "SupportedVersions":["1.0", "1.1"],
  "CurrentVersion":"1.1"
}
```

- GET-Request:
`https://bmsserver.customer.net/bConnect/version.xml`

Result:

```
<Version>
  <CurrentVersion>1.1</CurrentVersion>
  <SupportedVersions>
    <d2p1:string>1.0</d2p1:string>
    <d2p1:string>1.1</d2p1:string>
  </SupportedVersions>
</Version>
```

5.2 Controller Info

The *Info* controller provides actions concerning the *Info* object. *Info* is a bConnect internal object type which provides general information about the bConnect web server.

To request the *Info* object the URL must not contain version information and the provided user does not need configured rights in the baramundi Management Suite (bMS).

URL: https://<bmsserver>:<port>/bConnect/Info

Method	URL parameters	Description	Since bMS
GET	None	Returns an object of type <i>Info</i> .	2014

[Properties of Info.](#)

Examples:

- GET-Request:
https://bmsserver.customer.net/bConnect/info.json

Result:

```
{
  "Description": "Interface to baramundi Management Server",
  "Company": "baramundi software GmbH",
  "Copyright": "Copyright (c) baramundi software GmbH 2023",
  "bConnectVersion": "1.1",
  "bMSVersion": "23.1.xxx.x",
  "UserContext": "user@customer.net"
}
```
- GET-Request:
https://bmsserver.customer.net/bConnect/info.xml

Result:

```
<Info>
  <Company>baramundi software GmbH</Company>
  <Copyright>Copyright (c) baramundi software GmbH 2023</Copyright>
  <Description>Interface to baramundi Management Server</Description>
  <UserContext>user@customer.net</UserContext/>
  <bConnectVersion>1.1</bConnectVersion>
  <bMSVersion>23.1.xxx.x</bMSVersion>
</Info>
```

5.3 Controller Search

The *Search* controller provides actions to search over objects of the bMS. The search results are stored in *SearchResult* objects. Currently supported objects are of type *Endpoint*, *Job*, *OrgUnit*, *DynamicGroup*, *StaticGroup*, *Application* AD object and *App*.

The *Search* controller looks for org units only in nodes "Logical groups", "Dynamic groups", "Static groups", "Jobs", "Applications" and "Apps".

To request *SearchResult* objects data the URL must contain the bConnect version information. Only those objects can be scanned, the specified user has at least configured read rights on.

URL: https://<bmsserver>:<port>/bConnect/v1.1/Search

Method	URL parameters	Description	Since bMS
GET	Type=<object type> & Term=<search term>	Returns a list of <i>SearchResult</i> objects. <i>Type</i> specifies the objects to search over. Types "endpoint", "job", "orgunit", "group" (for static and dynamic groups), "adobject" and "software" (for applications and apps) are supported. <i>Term</i> specifies the search string (at least two characters) to look for in the objects of the specified type. To request the whole object the ID of the <i>SearchResult</i> object can be used. Works similar to the search in the bMC.	2014 R2

[Properties of Search.](#)

Examples:

- GET-Request:
`https://bmsserver.customer.net/bConnect/v1.1/search.json?type=endpoint&term=test`

Result:

```
[{
  "Id": "d9d4b3d9-8265-4001-ab23-993228a18b2f",
  "Name": "TestWindows",
  "AdditionalInfo": "",
  "Type": "1",
  "Deactivated": null
}]
```

5.4 Controller OrgUnits

The *OrgUnits* controller provides actions concerning the *OrgUnit* objects, the organizational unit objects, of the bMS.

To request *OrgUnit* objects data the URL must contain the bConnect version information. Only those *OrgUnit* objects the specified user has at least configured read rights on will be available.

Previously, only organizational units contained in the well-known org units *LogicalGroups* and *JobManagement* could be requested. In bMS 2016 *StaticGroups* and *DynamicGroups* as well as in bMS 21016R2 *Applications* and *Apps* were added to the scope of this controller. A request for other elements than these six root org units or its children will lead to a HTTP response code *NotFound* (e.g. a request with the *ParentGuid* of one of these six root org units).

To create an *OrgUnit* object with HTTP method POST the appropriate rights (create object, modify) must be given for the specified parent org unit to the specified user. To update an *OrgUnit* object with HTTP method PATCH the user needs the appropriate rights (modify) for the specified object and if *ParentId* is specified also for the new parent org unit.

URL: https://<bmsserver>:<port>/bConnect/v1.1/OrgUnits

Method	URL parameters	Description	Since bMS
GET	None	Returns a list of all available <i>OrgUnit</i> objects.	2014 R2
GET	OrgUnit=<org unit id>	Returns a list of <i>OrgUnit</i> objects, contained in the specified organizational unit. Sub units are not considered.	2014 R2
GET	ID=<org unit id>	Finds an <i>OrgUnit</i> object by ID.	2014 R2
POST	None (Mandatory and optional properties must be specified in the appropriate format in the HTTP body. In case of XML, the elements must be sorted alphabetically and the appropriate namespaces must be appended as attributes to the root element.)	Creates a new <i>OrgUnit</i> object according to the specified mandatory and optional data. Mandatory properties: - Name (unique in parent org unit) - ParentId (id of a org unit) Optional properties: - Comment - Extension (for org units of node <i>Logical Groups</i> only) See properties of OrgUnit for further information.	2016
PATCH	ID=<org unit id> (The properties that are to be updated must be specified in the appropriate format in the HTTP body. In case of XML, the elements must be sorted alphabetically and the appropriate namespaces must be appended as attributes to the root element.)	Updates the specified properties of the <i>OrgUnit</i> object by ID. Optional properties: - Name (unique in parent org unit) - ParentId (id of a org unit) - Comment - Extension See properties of OrgUnit for further information.	2016
DELETE	ID=<id>	Deletes an <i>OrgUnit</i> object by ID.	2016

[Properties of OrgUnit.](#)

Examples:

- GET-Request:
`https://bmssserver.customer.net/bConnect/v1.1/orgunits.json?id=a79851ff-3c9e-4973-b828-ced7c6a39f4b`

Result:

```
{
  "Id": "a79851ff-3c9e-4973-b828-ced7c6a39f4b",
  "ParentId": "b2ef5159-d8b8-4e9e-92fe-96889cec8491",
  "Name": "MyJobs ",
  "GuidParent": "b2ef5159-d8b8-4e9e-92fe-96889cec8491",
  "Comment": "",
  "HierarchyPath": "JobManagement\\MyJobs",
  "Extension": null
}
```

5.5 Controller DynamicGroups

The *DynamicGroups* controller provides actions concerning the *DynamicGroup* objects. Unlike *LogicalGroups*, *DynamicGroups* act as views, combining all endpoints of type *WindowsEndpoints* matching certain conditions defined by a SQL statement. One endpoint can be in several *DynamicGroups*.

To request *DynamicGroup* data, the URL must contain the bConnect version information. Only *DynamicGroup* objects with effective read-access for the specified user will be available with GET method.

To create a *DynamicGroup* object with HTTP method POST the appropriate rights (create object, modify) must be given for the specified parent org unit to the user. If the property *ParentId* is not specified the root node for dynamic groups is used.

To update a *DynamicGroup* object with HTTP method PATCH the user needs the appropriate rights (modify) for the specified object and if *ParentId* is specified also for the new parent org unit.

URL: https://<bmsserver>:<port>/bConnect/v1.1/DynamicGroups

Method	URL parameters	Description	Since bMS
GET	None	Returns a list of all available <i>DynamicGroup</i> objects.	2016
GET	OrgUnit=<org unit id>	Returns a list of <i>DynamicGroup</i> objects, contained in the specified organizational unit. Sub units are not considered.	2016
GET	ID=<dynamic group id>	Finds a <i>DynamicGroup</i> object by ID.	2016
POST	None (Mandatory and optional properties must be specified in the appropriate format in the HTTP body. In case of XML, the elements must be sorted alphabetically and the appropriate namespaces must be appended as attributes to the root element.)	Creates a new <i>DynamicGroup</i> object according to the specified mandatory and optional data. Mandatory properties: - Name (unique in parent org unit) - Statement (valid SQL) Optional properties: - ParentId (id of an org unit) - Comment See properties of DynamicGroup for further information.	2016
PATCH	ID=<dynamic group id> (The properties that are to be updated must be specified in the appropriate format in the HTTP body. In case of XML, the elements must be sorted alphabetically and the appropriate namespaces must be appended as attributes to the root element.)	Updates the specified properties of a <i>DynamicGroup</i> object by ID. Optional properties: - Name (unique in parent org unit) - Statement (valid SQL) - ParentId (id of an org unit) - Comment See properties of DynamicGroup for further information.	2016
DELETE	ID=<dynamic group id>	Deletes a <i>DynamicGroup</i> object by ID.	2016

[Properties of DynamicGroup.](#)

Examples:

- GET-Request:
<https://bmsserver.customer.net/bConnect/v1.1/DynamicGroups.json?id=643E6070-E31A-4DE2-8372-900965047B39>
Result:

```
{
  "Id": "643E6070-E31A-4DE2-8372-900965047B39",
  "ParentId": "BDE918DC-89C0-458A-92F7-0BB9147A2706",
  "Name": "64-bit systems",
  "Statement": " WHERE \r\n(\r\n (machine.arch =
                'x64')\r\n)\r\n",
  "Comment": ""
}
```
- POST-Request:
<https://bmsserver.customer.net /bConnect/v1.1/dynamicgroups.xml>
Body (xml format):

```
<DynamicGroup
xmlns:i="http://www.w3.org/2001/XMLSchema-instance"
xmlns="http://schemas.datacontract.org/2004/07/Baramundi.Bms.Server.Connector.Rest.BC
onnect.DataModel">
  <Comment>A group of all English systems.</Comment>
  <Name>English systems</Name>
  <ParentId>BDE918DC-89C0-458A-92F7-0BB9147A2706</ParentId>
  <Statement>WHERE ( machine.lcid = 1033 )</Statement>
</DynamicGroup>
```


Result:

```
<DynamicGroup xmlns:i="http://www.w3.org/2001/XMLSchema-instance"
xmlns="http://schemas.datacontract.org/2004/07/Baramundi.Bms.Server.Connector.Rest.BC
onnect.DataModel">
  <Comment>A group of all English systems.</Comment>
  <Id>D32CE263-AD72-4D86-B093-5796AC7F7D3F</Id>
  <Name>English systems</Name>
  <ParentId>BDE918DC-89C0-458A-92F7-0BB9147A2706</ParentId>
  <Statement>WHERE ( machine.lcid = 1033 )</Statement>
</DynamicGroup>
```
- PATCH-Request:
<https://bmsserver.customer.net /bConnect/v1.1/dynamicgroups?id=D32CE263-AD72-4D86-B093-5796AC7F7D3F>
Body (json format):

```
{
  "Name": "Systems with englisch lcid",
  "Comment": ""
}
```

Result:

```
{
  "Id": "41068A9F-405E-49BB-891B-65FAF1107CD9",
  "ParentId": "BDE918DC-89C0-458A-92F7-0BB9147A2706",
  "Name": "Systems with englisch lcid", "Statement": "WHERE ( machine.lcid = 1033 )",
  "Comment": ""
}
```

- DELETE-Request:
<https://bmsserver.customer.net /bConnect/v1.1/dynamicgroups.json?id=D32CE263-AD72-4D86-B093-5796AC7F7D3F>

Result: *No Content.*

5.6 Controller UniversalDynamicGroups

The *UniversalDynamicGroups* controller provides actions concerning the *UniversalDynamicGroup* objects. *UniversalDynamicGroups* act as views, combining endpoints of all types which match the configured conditions. One endpoint can be in several *UniversalDynamicGroups*.

To request *UniversalDynamicGroup* data, the URL must contain the bConnect version information. Only *UniversalDynamicGroup* objects with effective read-access for the specified user will be available with GET method.

URL: https://<bmsserver>:<port>/bConnect/v1.1/UniversalDynamicGroups

Method	URL parameters	Description	Since bMS
GET	None	Returns a list of all available <i>UniversalDynamicGroup</i> objects.	2020 R2
GET	OrgUnit=<org unit id>	Returns a list of <i>UniversalDynamicGroup</i> objects, contained in the specified organizational unit. Sub units are not considered.	2020 R2
GET	ID=<universal dynamic group id>	Finds a <i>UniversalDynamicGroup</i> object by ID.	2020 R2
	IsArgusSynced=<true / false>	Optional for all GET methods: If "true", only Universal Dynamic Groups synchronized with Argus Cockpit will be returned If "false", only Universal Dynamic Groups not synchronized with Argus Cockpit will be returned	2020 R2

[Properties of UniversalDynamicGroup.](#)

Examples:

- GET-Request:
`https://bmsserver.customer.net/bConnect/v1.1/UniversalDynamicGroups.json?id=643E6070-E31A-4DE2-8372-900965047B39`

Result:

```
{
  "Id": "643E6070-E31A-4DE2-8372-900965047B39",
  "ParentId": "BDE918DC-89C0-458A-92F7-0BB9147A2706",
  "Name": "64-bit systems",
  "DisplayName": "64-bit systems",
  "IsArgusSynced": "true",
  "Comment": "",
  "HierarchyPath": "Dynamic Groups"
}
```

5.7 Controller StaticGroups

The *StaticGroups* controller provides actions concerning the *StaticGroup* objects. Like *DynamicGroups*, *StaticGroups* act as views and can be used to manually group all type of endpoints outside of the logical group structure. One endpoint can be in several *StaticGroups*.

To request *StaticGroup* data, the URL must contain the bConnect version information. Only *StaticGroup* objects with effective read-access for the specified user will be available with GET method.

To create a *StaticGroup* object with HTTP method POST the appropriate rights (create object, modify) must be given for the specified parent org unit to the user. If property *ParentId* is not specified the root node for static groups is used.

To update a *StaticGroup* object with HTTP method PATCH the user needs the appropriate rights (modify) for the specified object and if *ParentId* is specified also for the new parent org unit.

URL: https://<bmsserver>:<port>/bConnect/v1.1/StaticGroups

Method	URL parameters	Description	Since bMS
GET	None	Returns a list of all available <i>StaticGroup</i> objects.	2016
GET	OrgUnit=<org unit id>	Returns a list of <i>StaticGroup</i> objects, contained in the specified organizational unit. Sub units are not considered.	2016
GET	ID=<static group id>	Finds a <i>StaticGroup</i> object by ID.	2016
POST	None (Mandatory and optional properties must be specified in the appropriate format in the HTTP body. In case of XML, the elements must be sorted alphabetically and the appropriate namespaces must be appended as attributes to the root element.)	Creates a new <i>StaticGroup</i> object according to the specified mandatory and optional data. Mandatory properties for <i>StaticGroups</i> : - Name (unique in parent org unit) Optional properties: - ParentId (id of an org unit) - EndpointIds - Comment See properties of StaticGroup for further information.	2016
PATCH	ID=<static group id> (The properties that are to be updated must be specified in the appropriate format in the HTTP body. In case of XML, the elements must be sorted alphabetically and the appropriate namespaces must be appended as attributes to the root element.)	Updates specified properties of a <i>StaticGroup</i> object by ID. Optional properties: - Name (unique in parent org unit) - ParentId (id of an org unit) - EndpointIds - Comment See properties of StaticGroup for further information.	2016
DELETE	ID=<static group id>	Deletes a <i>StaticGroup</i> object by ID.	2016

[Properties of StaticGroup.](#)

Examples:

- GET-Request:
https://bmsserver.customer.net/bConnect/v1.1/StaticGroups.json?id=41F24185-71DB-47F0-B036-05C5C2616479
Result:

```
{
  "Id": "41F24185-71DB-47F0-B036-05C5C2616479",
  "ParentId": "5020494B-04D3-4654-A256-80731E953746",
  "Name": "Server Group A",
  "Comment": "",
  "EndpointIds":
  [
    "4B3C0BBE-C7BC-4CF8-9598-0CD86DB3639A",
    "CC14E396-3919-4EDE-B302-D2FAD35B0986",
    "D13F5074-E773-4445-93F1-ACE372B54A21"
  ]
}
```
- POST-Request:
https://bmsserver.customer.net /bConnect/v1.1/staticgroups
Body (xml format):

```
<StaticGroup xmlns:i="http://www.w3.org/2001/XMLSchema-instance"
xmlns="http://schemas.datacontract.org/2004/07/Baramundi.Bms.Server.Connector.Rest.BC
onnect.DataModel">
  <Comment>A group of servers.</Comment>
  <EndpointIds xmlns:d3p1="http://schemas.microsoft.com/2003/10/
    Serialization/Arrays">
    <d3p1:string>CC28388A-5156-4BDE-A411-C2A5AD42EA9E</d3p1:string>
    <d3p1:string>37762ED0-9A20-4E44-8EA4-A97FF4625439</d3p1:string>
  </EndpointIds>
  <Name>Server Group B</Name>
  <ParentId>5020494B-04D3-4654-A256-80731E953746</ParentId>
</StaticGroup>
```


Result:

```
<StaticGroup xmlns:i="http://www.w3.org/2001/XMLSchema-instance"
xmlns="http://schemas.datacontract.org/2004/07/Baramundi.Bms.Server.Connector.Rest.BC
onnect.DataModel">
  <Comment>A group of servers.</Comment>
  <EndpointIds xmlns:d3p1="http://schemas.microsoft.com/2003/10/
    Serialization/Arrays">
    <d3p1:string>CC28388A-5156-4BDE-A411-C2A5AD42EA9E</d3p1:string>
    <d3p1:string>37762ED0-9A20-4E44-8EA4-A97FF4625439</d3p1:string>
  </EndpointIds>
  <Id>1B3B05B0-2DDD-4847-89CF-2E726BB918AE</Id>
  <Name>Server Group B</Name>
  <ParentId>5020494B-04D3-4654-A256-80731E953746</ParentId>
</StaticGroup>
```

- PATCH-Request:

<https://bmsserver.customer.net /bConnect/v1.1/staticgroups?id=1B3B05B0-2DDD-4847-89CF-2E726BB918AE>

Body (xml format):

```
<StaticGroup xmlns:i="http://www.w3.org/2001/XMLSchema-instance"
xmlns="http://schemas.datacontract.org/2004/07/Baramundi.Bms.Server.Connector.Rest.BC
onnect.DataModel">
  <Comment>Updated the name of this group.</Comment>
  <Name>Server Group NEW</Name>
</StaticGroup>
```

Result:

```
<StaticGroup xmlns:i="http://www.w3.org/2001/XMLSchema-instance"
xmlns="http://schemas.datacontract.org/2004/07/Baramundi.Bms.Server.Connector.Rest.BC
onnect.DataModel">
  <Comment>Updated the name of this group.</Comment>
  <EndpointIds xmlns:d3p1="http://schemas.microsoft.com/2003/10/
    Serialization/Arrays">
    <d3p1:string>CC28388A-5156-4BDE-A411-C2A5AD42EA9E</d3p1:string>
    <d3p1:string>37762ED0-9A20-4E44-8EA4-A97FF4625439</d3p1:string>
  </EndpointIds>
  <Id>1B3B05B0-2DDD-4847-89CF-2E726BB918AE</Id>
  <Name>Server Group NEW</Name>
  <ParentId>5020494B-04D3-4654-A256-80731E953746</ParentId>
</StaticGroup>
```

- DELETE-Request:

<https://bmsserver.customer.net /bConnect/v1.1/staticgroups.json?id=1B3B05B0-2DDD-4847-89CF-2E726BB918AE>

Result: *No Content.*

5.8 Controller Endpoints

The *Endpoints* controller provides actions concerning the *Endpoint* objects, the managed clients, of the bMS. All endpoint types, like *Windows*, *Network*, *Android*, *iOS* and *Mac* are supported. *Android*, *iOS* and *Mac* endpoints are of type *BmsNetEndpoint*.

To request *Endpoint* object data the URL must contain the bConnect version information. Only those *Endpoint* objects, the specified user has at least configured read rights on, will be available.

To create an *Endpoint* object with HTTP method POST in the specified organizational unit (OrgUnit) the appropriate rights (create object) must be given for the specified parent org unit to the user. To update an *Endpoint* object with HTTP method PATCH the user needs the appropriate rights (modify) for the specified object and if *ParentId* is specified also for the new parent org unit.

URL: https://<bmsserver>:<port>/bConnect/v1.1/Endpoints

Method	URL parameters	Description	Since bMS
GET	None	Returns a list of all available <i>Endpoint</i> objects.	2014
GET	OrgUnit=<org unit id>	Returns a list of <i>Endpoint</i> objects, contained in the specified organizational unit. Sub units are not considered.	2014 R2
GET	UniversalDynamicGroup=<universal dynamic group id>	Returns a list of <i>Endpoint</i> objects, contained in the specified universal dynamic group.	2020 R2
GET	DynamicGroup=<dynamic group id>	Returns a list of <i>Endpoint</i> objects, contained in the specified dynamic group.	2016
GET	StaticGroup=<static group id>	Returns a list of <i>Endpoint</i> objects, contained in the specified static group.	2016
GET	ID=<endpoint id>	Finds an <i>Endpoint</i> object by ID.	2014
GET	User=<user_principal_name>	Returns a list of <i>Endpoint</i> objects filtered by the specified "Registered User"	2018 R2
	PubKey=<true / false>	Optional for all GET methods: If "true", the returned objects will contain also the appropriate public key of the endpoint certificate (if stored for the client). Applies to endpoints of type <i>Windows</i> only.	2016
	InstalledSoftware=<true / false>	Optional for all GET methods: If "true", the returned endpoint objects will contain a list of the installed software with the bMS	2016 R2
	SnmpData=<true / false>	Optional for all GET methods: If "true", the returned endpoint objects will contain the appropriate SNMP data (if SNMP data are available)	2017R1
POST	None (Mandatory and optional properties must be specified in the appropriate format in the HTTP body. In case of XML, the	Creates a new <i>Endpoint</i> object according to the specified mandatory and optional data. Mandatory properties for all endpoint types: Type,	2014 R2

	elements must be sorted alphabetically and the appropriate namespaces must be appended as attributes to the root element.)	• DisplayName Optional properties for all endpoint types: • GuidOrgUnit (id of parent org unit) • Comments Optional for <i>WindowsEndpoint</i>: • HostName(15 chars, IETF RFC 952) (DisplayName by default) • Options (Format: decimal; When creating an endpoint, the settings for Automatic Usage Tracking (0x00000100) and Energy Management (0x00000200) cannot be set manually. The default settings from the database will be taken. Depending on their values, the default value for Options is one of 50, 306, 562 or 818.) • PrimaryMAC (M = one hex digit, Format: MM:MM:MM:MM:MM:MM) • Domain • GuidBootEnvironment • GuidHardwareProfile • Mode • ExtendedInternetMode • PrimaryUser (Principal or Down-Level Logon name (e.g. domain\user)) • PrimaryIP (primary IP-address of an endpoint) • PrimarySubnetMask • MicrosoftUpdateProfile • GuidMicrosoftUpdateProfile Optional for <i>MacEndpoint</i>: • HostName (is set after enrollment if empty) Optional for <i>BmsNetEndpoint</i>: • PrimaryUser (user's email address or down-level logon name (e.g. domain\user)) • Owner (default 0) • ComplianceCheckCategory (default 0) • AndroidEnterpriseProfileType (only valid for type Android) • ForceMobileDataOnEnrollment (only valid for type Android) • AppleManagementMode (only valid for type ios) • ManagedAppleId (only valid for ios and when AppleManagementMode is set to UserEnrollment) • EmailLanguageId • EmailRecipient Additional mandatory for <i>IndustrialEndpoint</i>:	
--	---	--	--

		- IndustrialData - PrimaryIP Additional mandatory for <i>NetworkEndpoint</i>: - PrimaryIP Optional for <i>IndustrialEndpoint</i>: - PrimaryMAC Optional for <i>NetworkEndpoint</i>: - PrimaryMAC See properties of Endpoint for further information. The returned object is of type NewEndpoint.	
PATCH	ID=<endpoint id> (The properties that are to be updated must be specified in the appropriate format in the HTTP body. In case of XML, the elements must be sorted alphabetically and the appropriate namespaces must be appended as attributes to the root element.)	Updates the specified properties of the <i>Endpoint</i> object by ID. Optional properties for all endpoint types: - DisplayName, - GuidOrgUnit (id of parent org unit) - Comments Optional for <i>WindowsEndpoint</i>: - HostName (15 chars, IETF RFC 952) - Options (Format: decimal) - PrimaryMAC (Format see POST) - Domain - GuidBootEnvironment - GuidHardwareProfile - PublicKey (without spaces) - Mode - ExtendedInternetMode - PrimaryUser (Principal or Down-Level Logon name (e.g. domain\user)) - PrimaryIP (primary IP-address of an endpoint) - PrimarySubnetMask - CustomStateText (a user defined string; prefixed and suffixed whitespaces are not supported) - CustomStateType (a user defined integer value) - MicrosoftUpdateProfile - GuidMicrosoftUpdateProfile Optional for <i>MacEndpoint</i>: - HostName Optional for <i>BmsNetEndpoint</i>: - PrimaryUser (users email-Address or Down-Level Logon name (e.g. domain\user)) - Owner (default 0) - ComplianceCheckCategory (default 0) - SerialNumber Optional for <i>IndustrialEndpoint</i>: - IndustrialData - PrimaryIP - PrimaryMAC	2016

		Optional for <i>NetworkEndpoint</i> : - PrimaryIP - PrimaryMAC See properties of Endpoint for further information. The returned object is of type <i>Endpoint</i>	
DELETE	ID=<endpoint id>	Deletes the <i>Endpoint</i> object by ID.	

[Properties of Endpoint.](#)

Examples:

- GET-Request:
`https://bmsserver.customer.net /bConnect/v1.1/endpoints.json?id=bd4a57c1-efe2-45bb-bf4a-58a5f4164d52`

Result:

```
{
  "Id":"bd4a57c1-efe2-45bb-bf4a-58a5f4164d52",
  "GuidGroup":"b2ef5159-d8b8-4e9e-92fe-96889cec8491",
  "DisplayName":"vm-cb-ws2008R2",
  "HostName":"vm-cb-ws2008R2",
  ...
}
```

- POST-Request:
`https://bmsserver.customer.net /bConnect/v1.1/endpoints.xml`

Body (xml format):

```
<Endpoint
xmlns:i="http://www.w3.org/2001/XMLSchema-instance"
xmlns="http://schemas.datacontract.org/2004/07/Baramundi.Bms.Server.Connector.Rest.BC
onnect.DataModel">
  <DisplayName>DeviceDisplayName</DisplayName>
  <GuidGroup>b2ef5159-d8b8-4e9e-92fe-96889cec8491</GuidGroup>
  <HostName>DeviceHostName</HostName>
  <Options>563</Options>
  <PrimaryMAC>12:34:56:78:90:12</PrimaryMAC>
  <Type>1</Type>
</Endpoint>
```

Result:

```
<NewEndpoint xmlns:i="..." xmlns="...">
  <Endpoint>
    <Activity i:nil="true"/>
    <DisplayName>DeviceDisplayName</DisplayName>
    ...
    <GuidGroup>B2EF5159-D8B8-4E9E-92FE-96889CEC8491</GuidGroup>
    <HostName>DeviceHostName</HostName>
    <Options>563</Options>
    <PrimaryMAC>12:34:56:78:90:12</PrimaryMAC>
    ...
    <Type>1</Type>
    ...
  </Endpoint>
```

```
<EnrollmentToken i:nil="true" />  
<TokenValidUntil i:nil="true" />  
</NewEndpoint>
```

- DELETE-Request:

<https://bmsserver.customer.net /bConnect/v1.1/endpoints.json?id=bd4a57c1-efe2-45bb-bf4a-58a5f4164d52>

Result: *No Content.*

5.9 Controller Jobs

The *Jobs* controller provides actions concerning Jobs for all platforms. Job objects can be of type *Windows*, *MobileAndMac* or *Universal* depending on the platform they are created for. Jobs of type *WindowsJob* are for endpoints of type *WindowsEndpoint*, Jobs of type *MobileAndMac* are for endpoints of type *AndroidEndpoint*, *iOSEndpoint* or *MacEndpoint* and *Universal* jobs are for *Industrial* endpoints.

The following URL allows you to request *Job* objects. You will only be able to retrieve Jobs which you are allowed to read according to your bMS security configuration.

URL: https://<bmsserver>:<port>/bConnect/v1.1/Jobs

Method	URL parameters	Description	Since bMS
GET	None	Returns a list of all available <i>Job</i> objects.	2014
GET	OrgUnit=<org unit id>	Returns a list of all available <i>Job</i> objects, contained in the specified organizational unit. Sub units are not considered.	2014
GET	ID=<job id>	Finds a <i>Job</i> object by ID.	2014
GET	User=<user_principal_name>	Returns a list of all released <i>Job</i> objects for the specified user	2018 R2
	Steps = <true / false>	Optional for all GET methods: If "true", the returned job objects will also contain the steps.	2016 R2
POST	None	Creates a new <i>Job</i> and returns it, including its <i>Job</i> steps Mandatory properties for Job: • Name • Type • List of Steps	2019
PATCH	ID=<job id>	Updates a set of properties of the <i>Job</i> with the specified ID. If the <i>Job</i> is assigned already, bConnect will return an error (BadRequest) as updating assigned <i>Jobs</i> may lead to inconsistencies.	2019
	ignoreAssignments=<true/false>	Optional for PATCH: If this parameter is set to true, bConnect will update the specified <i>Job</i> without error, even if it is assigned already.	2019
DELETE	ID=<job id>	Deletes the <i>Job</i> object by ID.	2019

[Properties of Job.](#)

Examples:

GET-Request:

<https://bmsserver.customer.net/bConnect/v1.1/jobs.json?id=104196a1-1862-45fe-be3f-b312fcfc1ca9>

Result:

```
{
  "Id": "104196a1-1862-45fe-be3f-b312fcfc1ca9",
  "ParentId": "C6567FDB-74B4-40C1-846D-12011A163B4A",
  "Type": "MobileAndMac",
  "Name": "Hallo",
  "Initiator": "initiator@test.net",
  "MobileAndMacProperties": {
    "CanRunOnMobileDevice": true,
    "CanRunOnMac": true,
    "RequiredComplianceLevel": "Unknown",
    "Repetition": 31449600
  }
}
```

PATCH-Request:

<https://bmsserver.customer.net/bConnect/v1.1/jobs.json?id=104196a1-1862-45fe-be3f-b312fcfc1ca9>

Request:

```
{
  "AbortOnError": true,
}
```

Result:

```
{
  "Id": "104196a1-1862-45fe-be3f-b312fcfc1ca9",
  "ParentId": "C6567FDB-74B4-40C1-846D-12011A163B4A",
  "Type": "MobileAndMac",
  "Initiator": "initiator@test.net",
  "Name": "Hallo",
  "AbortOnError": true,
  "MobileAndMacProperties": {
    "CanRunOnMobileDevice": true,
    "CanRunOnMac": true,
    "RequiredComplianceLevel": "Unknown",
    "Repetition": 31449600
  }
}
```

POST-Request:

<https://bmsserver.customer.net/bConnect/v1.1/jobs.json>

Request:

```
{
  "Id": "C745C8DB-642B-44D4-BED7-C5731BF2BD6D",
  "ParentId": "C6567FDB-74B4-40C1-846D-12011A163B4A",
  "Type": "Windows",
  "Name": "Install something",
  "DisplayName": "Install something",
  "AbortOnError": true,
  "Steps": [
    {
      "Applications": [
        {
          "Id": "503FE0F5-B631-48A5-9C73-D44ED43D3049"
        }
      ],
      "Type": "Deploy"
    }
  ]
}
```

Result:

```
{
  "Id": "C745C8DB-642B-44D4-BED7-C5731BF2BD6D",
  "ParentId": "C6567FDB-74B4-40C1-846D-12011A163B4A",
  "Type": "Windows",
  "Name": "Install something",
  "DisplayName": "Install something",
  "Initiator": "initiator@test.net",
  "AbortOnError": true,
  "Steps": [
    {
      "Applications": [
        {
          "Id": "503FE0F5-B631-48A5-9C73-D44ED43D3049",
          "Name": "AWDAWD AWDAWD"
        }
      ],
      "Sequence": 1,
      "Type": "Deploy"
    }
  ],
}
```

```
"WindowsProperties": {  
  "JobCount": -1,  
  "Priority": 500,  
  "Unlocksequence": "6355",  
  "MinBandwidth": 100000000,  
  "Options": {  
    "IgnoreBandwidth": true  
  },  
  "UserActionType": "CanNotInfluence",  
  "KeyMouseLockType": "DetermineAutomatically",  
  "PrePostInstallType": "ExecutePreAndPostInstallSteps",  
  "JobStartType": "Active",  
  "AtEndOfJobAction": "NoAdditionalAction",  
  "InfoWindowType": "DetermineAutomatically",  
  "MaxDelayMinutes": 90  
}  
}
```

DELETE-Request:

<https://bmsserver.customer.net/bConnect/v1.1/jobs.json?id=104196a1-1862-45fe-be3f-b312fcfc1ca9>

Result: No-Content

5.10 Controller JobInstances

The *JobInstances* controller provides actions concerning the *JobInstance* objects, formerly the job targets, of the bMS. Job instances for all endpoint types are supported.

To request *JobInstance* object data the URL must contain the bConnect version information. Only those *JobInstance* objects, the specified user has at least configured read rights on their *Endpoint*, will be available.

To create a *JobInstance* the specified user needs read rights for the specific *Endpoint* and *Job* objects in the bMS. Furthermore the user needs the right to assign the *Endpoint* object a job and the right to assign the specific *Job* object.

URL: https://<bmsserver>:<port>/bConnect/v1.1/JobInstances

Method	URL parameters	Description	Since bMS
GET	None	Returns a list of all available <i>JobInstance</i> objects.	2014
GET	EndpointID=<endpoint id>	Returns a list of all <i>JobInstance</i> objects of the specified endpoint.	2014 R2
GET	JobId=<job id>	Returns a list of all <i>JobInstance</i> objects of the specified job.	2015
GET	ID=<job instance id>	Finds a <i>JobInstance</i> object by ID.	2014
GET	EndpointID=<endpoint id> & JobID=<job id> & StartIfExists=<false/true> DEPRECATED (use POST instead)	Assigns the specified endpoint the specified job by creating a new <i>JobInstance</i> object, which is configured to be started as soon as possible. If this <i>JobInstance</i> object exists already it is returned and the parameter "StartIfExists" specifies with the value "true" to start it if it is not running yet. Not supported on <i>Endpoints</i> of type Industrial	2014
	Initiator=<name>	Optional for assignment: Sets assignment initiator to "<name> via <bConnectUser>" Example: MyThirdPartyTool via admin@domain.com	
GET	EndpointID=<endpoint id> & User=<user_principal_name>	Returns a list of all <i>JobInstance</i> objects of <i>Jobs</i> released for the specified user. User must be the <i>registered user</i> of the specified <i>Endpoint</i> .	2018R2
GET	TimePeriod=<seconds>	Returns a list of all <i>JobInstance</i> objects whose <i>TimeLastAction</i> is at most <i>TimePeriod</i> seconds ago.	2020
	Steps = <true / false>	Optional for all above GET methods: If <i>true</i> , the returned job objects will also contain the steps.	2016 R2
	IncludeJobDefinition= <true/false>	Optional for all above GET methods: If true, the returned <i>JobInstance</i> object	2020

		will also contain the according <i>JobDefinition</i> object	
GET	ID=<job instance id> & cmd=<start/stop/resume>	Starts, resumes or stops the specified <i>JobInstance</i> object, according to the specified command. To “start” a <i>JobInstance</i> object successfully, it must not have the state “Running”. To “stop” a <i>JobInstance</i> object successfully, it must not have the state “Finished” with success. Resuming a <i>JobInstance</i> object applies for endpoints of type <i>WindowsEndpoints</i> only. To “resume” a <i>JobInstance</i> object successfully, it must have one of the following states: - Not “Running” and flag “Resume” - “Finished” and flag “Warning” or “Error” - Flags “Rescheduled” and “Error” For endpoints of type <i>WindowsEndpoint</i> the commands “start” and “resume” will be effectively only if the baramundi Management Server is running.	2015
POST	None	Creates a new <i>JobInstance</i> object using the mandatory data. Mandatory properties are: • EndpointId • JobDefinitionId	2020
	StartIfExists=<false/true>	Optional for POST: If a <i>JobInstance</i> for the specified <i>Endpoint</i> and <i>Job</i> already exists and this parameter is set to true, the <i>JobInstance</i> will be started again.	2020
DELETE	ID=<job instance id>	Deletes a <i>JobInstance</i> object specified by ID. For endpoints of type <i>WindowsEndpoint</i> the <i>JobInstance</i> object must not have the state “Running”. For endpoints of type <i>BmsNetEndpoints</i> a running <i>JobInstance</i> object is canceled first and afterwards deleted.	2015

[Properties of JobInstance.](#)

Examples:

- GET-Request (start job instance):
<https://bmserver.customer.net/bConnect/v1.1/jobinstances?Id=EB6A02C3-BDCF-49EA-BD71-C12BB5C3561D&cmd=start>

Result: No data.

- POST-Request:

<https://bmsserver.customer.net/bConnect/v1.1/jobinstances>

Request:

```
{
  "EndpointId": "AAFF8188-A10A-436C-9AEB-92D88BB31CCA",
  "JobDefinitionId": "E4B87F22-F0D8-4F13-92AD-485044743329"
}
```

Result:

```
{
  "Id": "8DBF2A27-FE91-46C6-0000-0000000086A9C",
  "EndpointId": "AAFF8188-A10A-436C-9AEB-92D88BB31CCA",
  "EndpointName": "bConnect-Device",
  "JobDefinitionId": "E4B87F22-F0D8-4F13-92AD-485044743329",
  "JobDefinitionName": "Inventur",
  "TimeStart": "2020-02-24T09:48:12.8463917Z",
  "TimeLastAction": "2020-02-24T09:48:12.8463917Z",
  "StateText": "Job assigned",
  "Properties": {},
  "Initiator": "user@domain.net",
  "EndpointCategory": 7,
  "JobDefinitionDisplayName": ""
}
```

- DELETE-Request:

<https://bmsserver.customer.net/bConnect/v1.1/jobinstances?Id=EB6A02C3-BDCF-49EA-BD71-C12BB5C3561D>

Result: *No data.*

5.11 Controller HardwareProfiles

The *HardwareProfiles* controller provides actions concerning the available *HardwareProfile* objects defined in the bMS.

To request *HardwareProfile* objects the URL must contain the bConnect version information. Only those *HardwareProfile* objects can be read, the specified user has at least configured read rights on.

URL: https://<bmsserver>:<port>/bConnect/v1.1/HardwareProfiles

Method	URL parameters	Description	Since bMS
GET	None	Returns a list of all available <i>HardwareProfile</i> objects.	2016
GET	ID=<hardware profile id>	Finds a <i>HardwareProfile</i> object by ID.	2016

[Properties of *HardwareProfile*.](#)

5.12 Controller BootEnvironment

The *BootEnvironment* controller provides actions concerning the Boot Environments objects of the bMS.

To request *BootEnvironment* object data the URL must contain the bConnect version information. Only those *BootEnvironment* objects, the specified user has at least configured read rights on, will be available.

URL: https://<bmsserver>:<port>/bConnect/v1.1/BootEnvironment

Method	URL parameters	Description	Since bMS
GET	None	Returns a list of all available <i>Boot Environment</i> objects.	2016R2
GET	ID=<id>	Returns a single <i>Boot Environment</i> object with the specified ID.	2016R2

[Properties of *BootEnvironment*](#)

5.13 Controller Variables

The *Variables* controller provides actions concerning the *Variable* objects of all bMS objects that support variables. These *ObjectVariables* are certain instances of corresponding [VariableDefinition](#) objects which can be managed with the [controller VariableDefinitions](#).

To request *Variable* object data the URL must contain the bConnect version information. Only those *Variable* objects can be read, the specified user has at least configured read rights on the corresponding object in the bMS. To set a new value for a *Variable* object, the specified user needs modify rights on the corresponding object in the bMS.

With bMS 2015 job objects of type *WindowsJob*, org units in node *Logical Groups* and all endpoint objects supported reading and modification of their variables over bConnect. Since Version 2016R2 it is also possible to read and modify variables for *Software* (Windows Application only) and *HardwareProfile* objects.

URL: https://<bmsserver>:<port>/bConnect/v1.1/Variables

Method	URL parameters	Description	Since bMS
GET	Scope=<scope>	Returns a list of <i>ObjectVariables</i> of corresponding objects specified by <i>Scope</i> . One <i>ObjectVariables</i> object contains all variables of the specific bMS object. Currently supported scopes are "Device", "MobileDevice" (DEPRECATED), "OrgUnit", "Job", "Software", "HardwareProfile", "NetworkDevice", "ICDevice", "AndroidDevice" and "IOSDevice". (Check properties of ObjectVariables for more information)	2015
GET	Scope=<scope> & Category=<var category>	Returns a list of <i>ObjectVariables</i> of corresponding objects specified by <i>Scope</i> . Only the variables of the specified <i>Category</i> are supplied in the <i>ObjectVariables</i> objects.	2015
GET	Scope=<scope> & Category=<var category> & Name=<var name>	Returns a list of <i>ObjectVariables</i> of corresponding objects specified by <i>Scope</i> . Only the variables of the specified <i>Category</i> and <i>Name</i> are supplied in the <i>ObjectVariables</i> objects.	2015
GET	Scope=<scope> & ObjectId=<object id>	Returns an <i>ObjectVariables</i> object with the variables of the object specified by <i>ObjectId</i> . (e.g. ID of a WindowsEndpoint) The specified object must match with the specified scope for a successful result.	2015
GET	Scope=<scope> & ObjectId=<object id> & Category=<var category>	Returns an <i>ObjectVariables</i> object specified by <i>ObjectId</i> . (e.g. ID of a WindowsEndpoint) The specified object must match with the specified scope. Only variables with the specified <i>Category</i> will be returned.	2015

GET	Scope=<scope> & ObjectId=<object id> & Category=<var category> & Name=<var name>	Returns an <i>ObjectVariables</i> object specified by <i>ObjectId</i> . (e.g. the ID of a <i>WindowsEndpoint</i>) The specified object must match with the specified scope. Only variables with the specified <i>Category</i> and <i>Name</i> will be returned.	2015
PUT	None	Updates the specific Variable objects, specified by object id, category and name with the specified values. The specified parameters need to be passed in the body of the HTTP-Request. The data needs to be in the same format as returned by a HTTP-Request with Method GET. See properties of ObjectVariables for further information. The returned object is of type <i>ObjectVariables</i> containing all the variables of the specified object. The Property <i>UseDefault</i> is considered for endpoints of type <i>BmsNetEndpoint</i> , <i>ICDevice</i> , <i>NetworkDevice</i> , <i>AndroidDevice</i> , and <i>IOSDevice</i> only and is <i>false</i> by default.	2015

[Properties of ObjectVariables.](#)

Examples:

- GET-Request:
`https://bmsserver.customer.net/bConnect/v1.1/variables.xml?scope=device&objectid=AA61A522-6EE6-4623-A47C-7A35D270F0DF`

Result:

```
<ObjectVariables xmlns:i="http://www.w3.org/2001/XMLSchema-instance"
xmlns="http://schemas.datacontract.org/2004/07/Baramundi.Bms.Server.Connector.Rest.BC
onnect.DataModel">
  <ObjectId>AA60A522-6EE6-4623-A47C-7A35D270F0DF</ObjectId>
  <Scope>Device</Scope>
  <Variables>
    <Variable>
      <Category>Location</Category>
      <Name>Building</Name>
      <UseDefault>true</UseDefault>
      <Value>Building1</Value>
      <Type>String</Type>
    </Variable>
    ...
    <Variable>
      <Category> Location </Category>
      <Name>Room</Name>
      <Value>Laboratory</Value>
      <Type>String</Type>
```

```
</Variable>  
  </Variables>  
</ObjectVariables>
```

- PUT-Request:

https://bmsserver.customer.net/bConnect/v1.1/variables.xml

Body (xml format):

```
<ObjectVariables xmlns:i="http://www.w3.org/2001/XMLSchema-instance"
xmlns="http://schemas.datacontract.org/2004/07/Baramundi.Bms.Server.Connector.Rest.BC
onnect.DataModel">
  <ObjectId>AA60A522-6EE6-4623-A47C-7A35D270F0DF</ObjectId>
  <Scope>Device</Scope>
  <Variables>
    <Variable>
      <Category>Location</Category>
      <Name>Building</Name>
      <UseDefault>>false</UseDefault>
      <Value>OtherBuilding</Value>
    </Variable>
  </Variables>
</ObjectVariables>
```

Result:

```
<ObjectVariables xmlns:i="http://www.w3.org/2001/XMLSchema-instance"
xmlns="http://schemas.datacontract.org/2004/07/Baramundi.Bms.Server.Connector.Rest.BC
onnect.DataModel">
  <ObjectId>AA60A522-6EE6-4623-A47C-7A35D270F0DF</ObjectId>
  <Scope>Device</Scope>
  <Variables>
    <Variable>
      <Category>Location</Category>
      <Name>Building</Name>
      <UseDefault>>false</UseDefault>
      <Value>OtherBuilding</Value>
    </Variable>
    ...
    <Variable>
      <Category> Location </Category>
      <Name>Room</Name>
      <UseDefault>>false</UseDefault>
      <Value>Laboratory</Value>
    </Variable>
  </Variables>
</ObjectVariables>
```


5.14 Controller Apps

The *Apps* controller provides actions concerning the *App* objects, of the bMS. All sorts of apps for endpoints of type *BmsNetEndpoint* are supported.

To request *App* object data the URL must contain the bConnect version information. Only those *App* objects, the specified user has at least configured read rights on, will be available.

URL: https://<bmsserver>:<port>/bConnect/v1.1/Apps

Method	URL parameters	Description	Since bMS
GET	None	Returns a list of all available <i>App</i> objects.	2016R2
GET	OrgUnit=<org unit id>	Returns a list of <i>App</i> objects, contained in the specified organizational unit. Sub units are not considered.	2016R2
GET	EndpointID=<id>	Returns a list of <i>App</i> objects that are installed on the specified endpoint.	2016R2
GET	ID=<id>	Returns a single <i>App</i> objects by ID.	2016R2

[Properties of Apps.](#)

5.15 Controller Icons

The *Icons* controller provides actions concerning the icons of the available apps in the bMS. This means the apps to install on devices as well as the inventoried ones.

To request *Icons* object data the URL must contain the bConnect version information. Only those *Icon* objects, the specified user has at least configured read rights on the appropriate *App* or *InventoryApp* object, will be available.

URL: https://<bmsserver>:<port>/bConnect/v1.1/Icons

Method	URL parameters	Description	Since bMS
GET	None	Returns a list of all available <i>Icon</i> objects.	2016R2
GET	Scope=<scope>	Returns a list of all <i>Icon</i> objects depending on the chosen scope. If scope is "App", a list of <i>Icon</i> objects of all available and installable <i>App</i> objects will be returned. If scope is "Inventory", a list of <i>Icon</i> objects of inventoried app objects will be returned.	2016R2
GET	AppID=<app id>	Returns a single <i>Icon</i> object specified by an App ID.	2016R2

[Properties of Icons.](#)

5.16 Controller Applications

The *Applications* controller provides actions concerning the *Application* objects, of the bMS. These are the software products, including Managed Software products that are installed on endpoints of the type *WindowsEndpoint*.

To request *Application* object data the URL must contain the bConnect version information. Only those *Application* objects, the specified user has at least configured read rights on, will be available.

To create an *Application* object with HTTP method POST in the specified organizational unit (OrgUnit) the appropriate rights (create object) must be given for the specified parent org unit to the user. To update an *Application* object with HTTP method PATCH the user needs the appropriate rights (modify) for the specified object and if *ParentId* is specified also for the new parent org unit.

URL: https://<bmsserver>:<port>/bConnect/v1.1/Applications

Method	URL parameters	Description	Since bMS
GET	None	Returns a list of all available <i>Application</i> objects.	2016R2
GET	OrgUnit=<org unit id>	Returns a list of <i>Application</i> objects, contained in the specified organizational unit. Sub units are not considered.	2016R2
GET	EndpointID=<endpoint id>	Returns a list of <i>Application</i> objects that are installed with the bMS on the specified endpoint.	2016R2
GET	ID=<id>	Returns an <i>Application</i> object by ID.	2016R2
POST	None (Mandatory and optional properties must be specified in the appropriate format in the HTTP body. In case of XML, the elements must be sorted alphabetically and the appropriate namespaces must be appended as attributes to the root element.)	Creates a new <i>Application</i> object according to the specified mandatory and optional data. Mandatory properties: - Name - Vendor - ValidForOS See properties of Applications for further information.	2017
PATCH	ID=<application id> (The properties that are to be updated must be specified in the appropriate format in the HTTP body. In case of XML, the elements must be sorted alphabetically and the appropriate namespaces must be appended as attributes to the root element.)	Updates the specified properties of the <i>Application</i> object by ID. Optional properties for all Application types: - Name - Comment - Vendor - Category - ParentId - Version - ValidForOS - EnableAUT - Installation - Uninstallation - ConsistencyChecks	2017

		• Cost • Files • SecurityContext • Licenses • AUT • SoftwareDependencies • SoftwareLicenseActions • Parameters • Hashes See properties of Applications for further information. The returned object is of type Application	
DELETE	ID=<application id>	Deletes the <i>Application</i> object by ID.	2017

[Properties of Applications.](#)

Examples:

- POST-Request:
https://bmsserver.customer.net /bConnect/v1.1/applications.json

Body (json format):

```
{
  "Name": "Office Application",
  "Comment": "Alternative Office App",
  "Vendor": "fsf",
  "Category": "basic set",
  "ParentId": "C36331A7-259C-4544-BC0F-804BEE3C3D67",
  "Version": "1.2.3.4",
  "ValidForOS": [
    "WindowsXP_x64",
    "WindowsServer2003_x64",
    "WindowsVista_x64",
    "WindowsServer2008_x64",
    "Windows7_x64",
    "Windows8",
    "WindowsServer2012_x64",
    "Windows10_x64"
  ],
  "Installation": {
    "Command": "InstallCommand.exe",
    "Parameter": "/s",
    "Engine": "baramundi Deploy Script",
    "EngineFile": "{DIP}\\Scripts\\BDS\\ Office_App.bds ",
    "Options": {
      "AllowReinstall": false,
      "UsebBT": true,
      "VisibleExecution": "NeedsDesktop",
      "CopyLocally": false,
      "DisableInputDevices": true,
      "DontSetAsInstalled": true,
      "MapShare": false,
      "Target": "C:\\",
      "RebootBehaviour": "DeferrableReboot"
    }
  },
}
```

```
"UserSettings": {
  "ExecuteAtEveryLogin": false,
  "baramundiDeployScript": "{DIP}\\Scripts\\BDS\\Office_App_Ubds.bds",
  "ValidForInstallUser": false,
  "RunbDSVisible": false,
  "CopyScriptToClient": false
},
"SoftwareDependencies": [
  {
    "DependencyId": "B9E90391-AECC-44D5-8FF6-3AFCD4F0683E",
    "DependencyType": "InstallBeforeIfNotInstalled",
    "ValidForOS": [
      "Windows7_x64",
      "WindowsServer2008R2_x64",
      "Windows8_x64",
      "WindowsServer2012_x64",
      "Windows10_x64",
      "WindowsServer2016_x64"
    ]
  }
],
"SecurityContext": "LocalInstallUser",
"Parameters": [
  {
    "Name": "MaxParralelUsers",
    "Type": "Number",
    "Value": "2",
    "DefaultValue": "3"
  }
]
}
```

5.17 Controller InventoryDataRegistryScans

The *InventoryDataRegistryScans* controller provides actions concerning *Inventory* objects with data of a registry scan. This data is gathered in the bMS with a Software scan.

To request *Inventory* objects with registry scan data the URL must contain the bConnect version information. Only those *Inventory* objects can be read, the specified user has at least configured read rights on the node *Inventoried Files* and the corresponding endpoint in the bMS. To delete *Inventory* objects the specified user needs read rights on the node *Inventoried Files* as well as read and modify rights on the endpoint. Inventory Registry scan data can be managed for endpoints of type *WindowsEndpoint* only.

URL: https://<bmsserver>:<port>/bConnect/v1.1/InventoryDataRegistryScans

Method	URL parameters	Description	Since bMS
GET	None	Returns a list of all available <i>Inventory</i> objects with the registry scan data.	2015
GET	EndpointID=<endpoint id>	Returns the <i>Inventory</i> object with the registry scan data of the endpoint with the specified ID.	2015
DELETE	EndpointID=<endpoint id>	Deletes all registry scan data of the endpoint with the specified ID.	2015

[Properties of *Inventory*.](#)

[Properties of *DataRegistry*.](#)